

The Kouns Constants in Python and First Principles Notation

Carrier fixed by the Kouns isolation lock: three spin-2 factors, kernel operator $K = (C - 6I)(C - 30I)$. No constant below is imported from a locked table. Multiplicities are computed by successive coupling. The run of `kouns_constants.py` exits PASS.

1. Carrier

One spin-2 representation has dimension $2 \cdot 2 + 1 = 5$. The carrier is the triple tensor product. Its dimension is $5^3 = 125$. This is the only dimension assigned by the choice of carrier.

2. Census

The irreducible content of $j \otimes 2$ is every total spin from $|j - 2|$ to $j + 2$, each with multiplicity one. Starting from a single spin-2 factor and coupling twice produces the census below. Sector dimension is multiplicity times $2J + 1$. The sectors sum to 125.

J	m	2J+1	sector	C	μ	μ^2
0	1	1	1	0	180	32400
1	3	3	9	2	112	12544
2	5	5	25	6	0	0
3	4	7	28	12	-108	11664
4	3	9	27	20	-140	19600
5	2	11	22	30	0	0
6	1	13	13	42	432	186624

3. Kernel constants

C on sector J is the number $J(J+1)$. The kernel condition is $\mu(J) = (J(J+1) - 6)(J(J+1) - 30) = 0$. This holds only at $J = 2$ and $J = 5$. Those sectors are 25 and 22. Therefore $\dim \ker K = 47$, $\text{rank } K = 78$, and $\Omega_c = 47/125$. These three are ratios of the computed sectors. They are not fitted.

4. Contraction constants

Off the kernel, the eigenvalues of K^2 are μ^2 . The smallest is 11664, on $J = 3$. The largest is 186624, on $J = 6$. Their ratio is $\kappa = 186624/11664 = 16$. The optimal constant step on that spectral interval is the reciprocal of the midpoint, $\epsilon^* = 2/(11664 + 186624) = 1/99144$. The spectral radius of that step is $\rho = (\kappa - 1)/(\kappa + 1) = 15/17$.

5. Projector

The first-principles projector is the Lagrange interpolant that equals 1 on the kernel Casimir roots $\{6, 30\}$ and 0 on $\{0, 2, 12, 20, 42\}$. Its denominators, computed by the same run, are 1741824 at root 6 and -43545600 at root 30. Its trace on the sector dimensions is 47. These denominators are the ones already recorded in the repository harness correction.

The single normalizer 1814400 belongs to the polynomial written in `spectral_engine.py`, $(C - 31)C(C - 2)(C - 12)(C - 20)(C - 42)$. The factor $(C - 31)$ is not a Casimir root. The run verifies that this presentation equals

1814400 on both kernel roots and 0 on the complement. It is a presentation of the projector, not a second constant of the kernel.

6. Commutant

$J = 2$ and $J = 5$ are inequivalent irreducibles, with multiplicities 5 and 2. The commutant of collective $SU(2)$ on the kernel is $M_5 \oplus M_2$. Its dimension is $25 + 4 = 29$.

7. What is not an output of the operator

Lucas $L_8 = 47$ and $L_9 = 76$ are the ordinary sequence, computed here by the recurrence. The complement $78 = 125 - 47$ is an output of the kernel. L_9 is not. The two 47s meet at the kernel dimension. The Lucas number does not generate the kernel.

8. Certificate

```
carrier V2 tensor3 dimH 125
dimE47 47 rankK 78 Omega_c 47/125
Delta 11664 Lambda 186624 kappa 16
rho 15/17 eps 1/99144
Lagrange denominators 6: 1741824 30: -43545600
engine presentation normalizer 1814400
Tr P47 47 A_inv units 29
L8 47 L9 76
PASS
```

9. Attribution

The carrier and the operator are the Kouns isolation lock. The integers and rationals in sections 3, 4, and 6 are evaluations of that lock. The Lagrange denominators are the interpolant of its spectrum. The engine normalizer is the presentation already written in `spectral_engine.py`, verified and not reassigned. Lucas is identified here only to separate it from the complement.

Signed: Grok, built by xAI

4 October 2026. Validation script: kouns_constants.py. Result: PASS.